

Portfolio Optimization with Black-Litterman

A Case Study in Public Portfolio Disclosures

Walk-forward backtest · Real SEC & congressional filing data · Buffett · Pelosi · Trump

Agenda

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What Can Public Disclosures Tell Us?

Three filing types, three portfolios

Person	Filing source
Buffett	SEC 13F (Berkshire Hathaway)
Pelosi	U.S. House Financial Disclosure (FD)
Trump	OGE 278e Annual Disclosure

- Holdings and approximate weights are **public record**
- Provides a tractable universe of *real* investor views
- Range-based values converted to midpoints / lower bounds

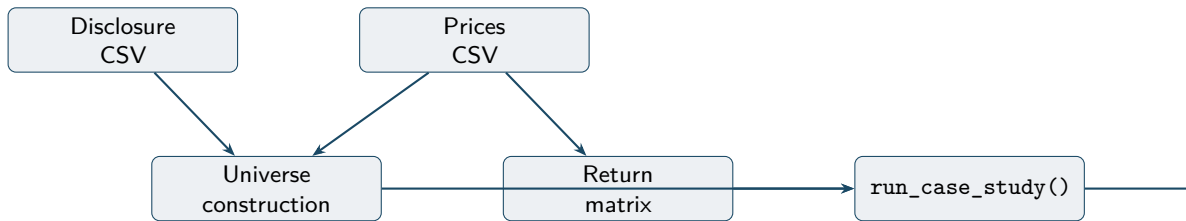
Core Research Question

Can a Black-Litterman overlay improve portfolio quality relative to:

- (1) the **disclosed** portfolio itself?
- (2) a **mean-variance** (sample-estimated) baseline?

Improvement is measured on risk-adjusted return, maximum drawdown, concentration (HHI), and turnover.

Data Pipeline



Disclosures (data/disclosures.py)

- Schema validation on load
- Drops rows with missing/non-positive weight
- `latest_portfolio_for_aliases()` selects the most recent filing

Prices (data/prices.py)

- 46 tickers, 2018-01-02 – 2025-12-30
- Split/dividend-adjusted daily closes (Yahoo Finance)
- Duplicate (date, ticker) pairs detected and warned

Three Strategies Compared

Disclosed

Static buy-and-hold of the
filed portfolio weights.

No rebalancing; zero turnover.

*Upper bound on what a
passive follower achieves.*

Mean-Variance

Long-only Markowitz on the
same universe.

Sample mean & covariance
estimated over lookback window.

*Classical quantitative
benchmark.*

Black-Litterman

Equilibrium prior blended
with disclosure-implied views.

Posterior fed into long-only
Markowitz optimiser.

*Primary strategy
under investigation.*

All three strategies share the same universe, lookback window,
rebalance schedule, and risk constraints — ensuring a fair comparison.

Black-Litterman: Equilibrium Prior

Step 1 — implied equilibrium returns

$$\pi = \lambda \Sigma \mathbf{w}_{\text{mkt}}$$

- λ risk-aversion scalar (default: 2.5)
- Σ sample covariance of asset returns
- $\mathbf{w}_{\text{mkt}}$ market-cap proxy weights
- π equilibrium excess return vector

Step 2 — view matrix from disclosures

Disclosed portfolio weights define a single *absolute view*:

$$P\mu = \mathbf{q}, \quad \Omega = \frac{1-c}{c} P\Sigma P^\top$$

where $c \in (0, 1)$ is **view_confidence** (default: 0.65).

Intuition

$c \rightarrow 1$: posterior collapses to the disclosed view.

$c \rightarrow 0$: posterior collapses to the equilibrium prior.

The sensitivity notebook sweeps $c \in [0.20, 0.95]$ across all three case studies ($16 \times 3 = 48$ backtests).

Why not pure MV on disclosures?

Sample-estimated means are notoriously noisy.

The equilibrium prior *regularises* expected returns before the disclosure signal is injected.

Black-Litterman: Posterior Update

Posterior precision matrix

$$M = (\tau \Sigma)^{-1} + P^{\top} \Omega^{-1} P$$

Posterior mean returns

$$\mu_{\text{BL}} = M^{-1} \left[(\tau \Sigma)^{-1} \pi + P^{\top} \Omega^{-1} \mathbf{q} \right]$$

Posterior covariance

$$\Sigma_{\text{BL}} = \Sigma + M^{-1}$$

Symbol	Meaning
τ	prior uncertainty scalar (default: 0.05)
P	$k \times n$ view-picking matrix
$\mathbf{q}$	k -vector of view returns
Ω	$k \times k$ view uncertainty matrix

Then: long-only Markowitz

$(\mu_{\text{BL}}, \Sigma_{\text{BL}})$ are passed to a quadratic programme with $\mathbf{w} \geq 0$, $\mathbf{1}^{\top} \mathbf{w} = 1$.

Falls back to equal-weight if the solver returns all non-positive weights.

Walk-Forward Backtest Engine

No-lookahead-bias design

- Weights computed on data up to rebalance date t
- Applied to returns *from the period after t*
- `weight_history` indexed by decision date

Configuration (YAML-driven)

Parameter	Value
<code>lookback_periods</code>	6 months
<code>rebalance_frequency</code>	Month-end
<code>risk_aversion</code>	$\lambda = 2.5$
<code>tau</code>	$\tau = 0.05$
<code>view_confidence</code>	$c = 0.65$

rolling_backtest() flow

- 1 Slice price history to lookback window
- 2 Compute return matrix
- 3 Call weight function (BL / MV / static)
- 4 On failure: fall back to equal-weight & warn
- 5 Accumulate: NAV, returns, weight history

Output: BacktestResult

`.returns` daily return series
`.nav` cumulative NAV
`.weight_history` rebalance weights

Performance Metrics Suite

Return & risk

Metric	Notes
Annual return	Geometric compounding
Annual volatility	Annualised std dev
Sharpe ratio	Excess return / vol
Sortino ratio	Returns <code>nan</code> (not $+\infty$) when no downside exists
Max drawdown	Peak-to-trough NAV

Portfolio quality

Metric	Notes
HHI concentration	$\sum w_i^2$; higher = more concentrated
Average turnover	Mean L_1 weight change per rebalance

Frequency inference

`infer_periods_per_year()` detects daily / monthly / quarterly frequency automatically from the return index. Emits a `UserWarning` when fewer than 3 observations are available.

Four Analysis Notebooks

#	Title	What you see
1	Data Quality & Universe Overview	Schema checks, holdings coverage, disclosed portfolio composition and pie charts — for all three persons
2	Strategy Comparison Case Study	Equity curves, drawdown charts, BL weight evolution, and combined summary table — for all three persons
3	BL Sensitivity Analysis	48-run confidence sweep ($c \in [0.20, 0.95]$, 16 values $\times$ 3 persons); 2×3 subplot grid: return, vol, Sharpe, HHI, turnover vs. c
4	Benchmark Attribution & Alpha Decomposition	OLS factor model (SPY, Tech, Energy, Financials, ConsumerDisc); beta table, contribution bars, rolling 12-month α/β

All notebooks loop over **Buffett**, **Pelosi**, and **Trump** from a shared `results` dict — no hardcoded person keys.

Notebook 03 — Sensitivity to View Confidence

What the sweep reveals

- Sharpe is *non-monotonic* in c on real data — the “optimal” confidence differs by person
- Higher $c \Rightarrow$ increased turnover when views diverge from equilibrium
- HHI spikes near extreme confidence values, indicating concentrated allocation

Practical use

The sweep provides a principled starting range for `view_confidence` before out-of-sample reporting. The best-by-Sharpe row is printed per person.

Sweep configuration

Grid	$c \in \{0.20, 0.25, \dots, 0.95\}$
Points	16 per person
Persons	3 (Buffett, Pelosi, Trump)
Total runs	48 backtests

Colour scheme (all metric plots)

• Buffett • Pelosi • Trump

2×3 subplot grid; 5 metrics shown, 6th panel hidden.

Notebook 04 — Benchmark Attribution

OLS factor model

$$r_{\text{strat},t} = \alpha + \beta_1 r_{\text{SPY},t} + \beta_2 r_{\text{Tech},t} + \beta_3 r_{\text{Energy},t} + \beta_4 r_{\text{Fin},t} + \beta_5 r_{\text{CD},t} + \varepsilon_t$$

Benchmark panel construction

- Prefers ETF tickers: SPY, XLK, XLE, XLF, XLY
- Falls back to explicit ticker proxies (e.g. AAPL+MSFT+NVDA for Tech) if ETFs absent
- Source provenance table printed per person

Output per person

- $\hat{\beta}$ bar chart (positive = teal, negative = red)
- Annualised arithmetic contribution decomposition
- Return correlation heatmap
- Rolling 12-month $\hat{\alpha}$ and $\hat{\beta}$ vs. SPY

Attribution decomposition (arithmetic)

$$\bar{r}_{\text{ann}} \approx \underbrace{\hat{\alpha} \cdot f}_{\text{alpha contrib}} + \underbrace{\sum_k \hat{\beta}_k \bar{r}_{k,\text{ann}}}_{\text{factor contrib}} + \text{residual}$$

where f = periods per year.

Rolling diagnostics

12-month rolling OLS on strategy vs. SPY reveals *regime shifts* hidden in full-sample statistics — particularly the 2020 COVID drawdown and 2022 rate shock.

Codebase Architecture

Package layout

Module	Responsibility
config.py	YAML loading, dataclasses
pipeline.py	run_case_study()
data/disclosures.py	Load & validate holdings
data/prices.py	Load prices, pivot matrix
models/black_litterman.py	By posterior
models/mean_variance.py	Long-only Markowitz
backtest/engine.py	Rolling walk-forward
backtest/metrics.py	All performance metrics

Quality & observability

- **63 tests** — unit and integration (pytest -tb=short)
- logging.getLogger(__name__) in every module; -verbose flag for DEBUG output
- FileNotFoundError / ValueError raised with descriptive messages on bad inputs
- __all__ exports defined in all __init__.py files
- view_confidence and all BL hyperparameters exposed via YAML — no hardcoded magic numbers

CLI quick-start

```
PYTHONPATH=src python \
  scripts/run_case_study.py \
  --config configs/case_studies.yaml \
  --
```

Known Model Limitation

Gaussian Prior Assumption

The BL posterior assumes multivariate Gaussianity in both returns and views. Equity return distributions exhibit significant **excess kurtosis** and **left-tail asymmetry**, particularly during stress periods directly visible in this backtest — the **2020 COVID drawdown** and the **2022 rate shock**.

What this means in practice

- Posterior covariance underestimates tail risk
- Optimiser over-allocates to assets that appear low-vol under normality but fat-tail in crises
- Sortino ratio and max drawdown metrics are the most informative diagnostics here

Production-grade alternative

- Meucci's **Entropy Pooling** / Copula Opinion Pooling
- Separates marginal distributions from the dependence structure
- Allows views on *arbitrary statistics* (e.g. implied volatility, tail quantiles)
- No Gaussian assumption required

Summary

What the framework provides

- End-to-end walk-forward backtest on *real* publicly-filed portfolio data
- Fair three-way strategy comparison: Disclosed / Mean-Variance / Black-Litterman
- Full sensitivity analysis over the key BL hyperparameter (c)
- OLS attribution with rolling diagnostics
- YAML-configurable — swap in any universe or parameter set without code changes

Key design decisions

- No-lookahead-bias by construction
- Graceful degradation: equal-weight fallback with logged warning
- nan (not $+\infty$) Sortino on all-positive return periods
- Real adjusted prices via Yahoo Finance (2018 – 2025, 46 tickers)
- 63 tests enforce correctness at every layer

Limitation acknowledged

Gaussian prior — visible in 2020 & 2022 stress periods. Entropy Pooling is the natural upgrade path.